💡 LIGHT – REFLECTION & REFRACTION

TOPPER NOTES

⚡ Physics Chapter 9/10 | 🔭 Ray Optics Masterclass | 🌻 Hinglish Edition

1. Reflection of Light & Laws of Reflection 🪞

Definition: The bouncing back of light into the same medium when it strikes a polished surface is called **Reflection**.

Laws of Reflection:

- **Rule 1:** Angle of incidence ($\angle i$) is always equal to the angle of reflection ($\angle r$). i.e., $\angle i = \angle r$.
- **Rule 2:** The incident ray, normal to the mirror at the point of incidence, and reflected ray all lie in the same plane.

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Text-Based Diagram (Laws of Reflection): Normal (N) | ^ | / \ | / \ Reflected Ray (r) Incident Ray |/ \ (i) / \ \ / \ \ / \ ═══════════╨═══╨═══════════════════╨═══════════ (Mirror Surface) Angle i = Angle r
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2. Spherical Mirrors (Goliya Darpan) 🌐

Spherical mirrors ke do types hote hain:

- **Concave Mirror (Avatal Darpan):** Reflecting surface inwards curved hoti hai (Converging mirror).
- **Convex Mirror (Uttal Darpan):** Reflecting surface outwards bulged hoti hai (Diverging mirror).

3. Sign Convention for Spherical Mirrors 📐

- Object hamesha mirror ke left mein rakha jata hai (Incident ray left se right jati hai).
- Saari distances pole (P) se measure hoti hain.
- Right side ki distances **Positive (+ve)** aur left side ki **Negative (-ve)** hoti hain.
- Principal axis ke upar height **+ve** (upright image) aur niche **-ve** (inverted image) hoti hai.

⭐ **Focal Length Rule:** Concave mirror ki focal length (f) hamesha **Negative (-ve)** hoti hai, aur Convex mirror ki **Positive (+ve)** hoti hai!

4. Refraction of Light (Prakash ka Apvartan) 🌊

Definition: Light jab ek transparent medium se dusre transparent medium mein jati hai, toh apne path se bend ho jati hai. Ise **Refraction** kehte hain.

Laws of Refraction (Snell's Law):

- **Rule 1:** Incident ray, refracted ray, and normal at the point of incidence all lie in the same plane.
- **Rule 2:** The ratio of sine of angle of incidence to the sine of angle of refraction is constant for a given pair of media.

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Formula: $\frac{\sin i}{\sin r} = \text{constant} = n_{21}$ (Refractive Index)
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💡 **Real-life Example:** Paani ke glass mein pencil ka tedha dikhna (bending of stick) aur paani ke talab ka shallow (kam gehra) lagna refraction ki wajah se hota hai.

5. Lenses (Lens) 🔍

- **Convex Lens (Uttal Lens):** Bich mein mota aur kinaron par patla hota hai (Converging lens). Real and inverted images banata hai mostly.
- **Concave Lens (Avatal Lens):** Bich mein patla aur kinaron par mota hota hai (Diverging lens). Always virtual and erect images banata hai.

6. Power of a Lens (Lens ki Shakti) ⚡

Definition: The degree of convergence or divergence of light rays achieved by a lens is called its power. It is reciprocal of focal length (in meters).

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Formula: $P = \frac{1}{f \text{ (in meters)}}$ OR $P = \frac{100}{f \text{ (in cm)}}$ Unit of Power: Dioptre (D)
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⭐ **Sign Rule for Power:** Convex lens ki power **Positive (+ve)** hoti hai, aur Concave lens ki power **Negative (-ve)** hoti hai!

7. 🌟 Dedicated Formula Corner (All Formulas at a Glance)

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1. Radius & Focal Length: R = 2f or f = R/2 2. Mirror Formula: 1/f = 1/v + 1/u 3. Lens Formula: 1/f = 1/v - 1/u 4. Magnification (Mirror): m = -v/u = h'/h 5. Magnification (Lens): m = v/u = h'/h 6. Refractive Index: n = c / v (speed of light in vacuum / medium) 7. Power of Lens: P = 1 / f(in meters)
```

8. Exam Short Tricks & Tips 🚀

- **Mirror vs Lens Formula confusion:** Mirror formula mein **PLUS (+)** sign hota hai ($1/v + 1/u$), jabki Lens formula mein **MINUS (-)** sign hota hai ($1/v - 1/u$).
- **Magnification Sign:** Agar m negative hai, toh image **Real & Inverted** hai. Agar m positive hai, toh image **Virtual & Erect** hai.

9. ⚡ Quick Revision Checklist (Last-Minute Summary) ✅

- [ ] **Laws of Reflection:** $\angle i = \angle r$.
- [ ] **Concave Mirror:** Real/Inverted images, negative focal length.
- [ ] **Convex Mirror:** Virtual/Erect smaller images, positive focal length.
- [ ] **Snell's Law:** $\sin i / \sin r = \text{constant}$.
- [ ] **Refractive Index:** $n = c/v$ (no units).
- [ ] **Lens Power:** Measured in Dioptre (D); $P = 1/f$.

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